HNCA: Hierarchical Neural Constraint Architecture — Cognition as Constraint Satisfaction across Two Coupled Scales

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> *"The engine is not searching for an answer. It is integrating differential equations, and the answer is where the energy stops."*

Abstract

We specify a two-scale substrate for cognition in which computation is constraint satisfaction rather than search or prediction. The **micro-layer** is a continuous-state cellular grid whose cells carry physical variables and whose edges are differential equations rather than learned weights. The **macro-layer** wraps clusters of those cells into regions with distinct natural frequencies, coupled by the Kuramoto model, reflecting the observation that brain regions operate at different characteristic rhythms rather than as one homogeneous field.

Processing has four stages: a perturbation is injected as a continuous stimulus; the injected energy breaks the grid's equilibrium; the engine integrates the constraint equations until the field settles; and the resulting topological pattern **is** the thought, not a code for one. No token is involved at any stage.

The macro-layer of this proposal is implemented and load-bearing in the current engine. The micro-layer is not, and §6 states precisely which part survived and why.

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1. Position

Two architectural options are commonly assumed to exhaust the space: a flat cellular grid, or a Transformer over lexical tokens. Neither is adequate for constraint-based cognition.

A **flat grid** has no notion of region. Every cell obeys the same dynamics at the same characteristic rate, so there is no mechanism by which one part of the system can operate on a different timescale from another — and differential timescale is what allows slow context to modulate fast processing.

A **token Transformer** operates on discrete lexical units and computes a distribution over continuations. Its geometry is a fully-connected attention matrix, which is to say it has no geometry: every unit is equidistant from every other, and spatial locality carries no meaning.

What is required instead is a **physics engine for cognition** — a system with local interaction rules, spatial structure, and multiple characteristic frequencies.

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2. The Micro-Layer: Continuous-State Cellular Dynamics

The foundational layer is a multi-dimensional continuous grid. A cell does not contain a string or an embedding; it represents a micro-column and carries a state vector of physical quantities: membrane potential V_m, a recovery variable u, and a local field potential.

The edges are **differential equations**, not learned parameters. The cell update implements a spiking model natively — for instance the Izhikevich formulation:

$$\frac{dV}{dt} = 0.04V^2 + 5V + 140 - u + I$$

$$\frac{du}{dt} = a(bV - u)$$

The significance is what is *absent*. There is no embedding table, no attention, and no parameter fitted by gradient descent. The constraints are stated, not learned, and the dynamics they induce are the computation.

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3. The Macro-Layer: Hierarchical Oscillators

A brain is not a homogeneous grid. Cortex, thalamus and amygdala operate at different characteristic frequencies, and that heterogeneity is functional rather than incidental.

Clusters of micro-cells are therefore wrapped into **macro-nodes**, each with a natural frequency ω_i, and coupled through the Kuramoto model. The engine continuously solves for phase across the network:

$$\frac{d\theta_i}{dt} \;=\; \omega_i \;+\; \frac{K}{N}\sum_{j=1}^{N}\sin(\theta_j - \theta_i)$$

Here ω_i is the natural frequency of a region and K is the coupling strength — the analogue of the white-matter tracts connecting regions.

Two properties follow. Regions with similar ω **entrain** easily and exchange influence readily. Regions with dissimilar ω resist entrainment, so a frequency separation acts as a *soft domain*

boundary requiring no gating logic and no explicit routing table. Domain isolation becomes a physical property rather than a control structure.

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4. The Constraint Engine

Input and output are both physical events.

Stage 1 — Inject perturbation. Rather than passing text to an encoder, a continuous stimulus (a current I) is injected into specific sensory cells.

Stage 2 — Disequilibrium. The injected energy breaks the grid's equilibrium. Kuramoto oscillators fall out of phase and cell states spike. The system is *forced* to react; nothing decides to respond.

Stage 3 — Continuous constraint solving. The integration loop runs. The engine is not searching a space of candidate answers — it is integrating differential equations across the edges. Energy propagates bound strictly by the local rules.

Stage 4 — Attractor convergence. The field settles into a new stable configuration. **This topological pattern is the thought.** A downstream decoder translates the settled state into text or action, and that translation is a separate, explicitly-named cost rather than part of the cognition.

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5. Comparison

Property	Token Transformer	Constraint engine
Fundamental unit	lexical tokens	continuous state variables
Mechanics	probabilistic next-symbol prediction	deterministic differential equations
Network geometry	fully-connected attention	spatially bounded topological grid
Output generation	sampling from a distribution	relaxation into an attractor
Timescales	one	many, set by ω_i per region

The intended consequence is a change in what is being programmed. One does not specify the logic; one specifies the **laws** the substrate obeys, and the logic is whatever those laws force.

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6. What Was Implemented, and What Was Not

This section is the paper's accounting against the running system, and it corrects an earlier presentation in which the whole proposal was described as live.

The macro-layer is implemented and load-bearing. Macro-nodes with per-region natural frequencies, coupled by the Kuramoto update, are in the current substrate (`struct MacroNode`, `physics.hpp`), and the frequency-separation-as-domain-boundary property of §3 carries through to the sub-band ladder of the VGHT layer. The claim that dissimilar ω resists entrainment and thereby isolates domains was tested and holds.

The micro-layer is not implemented. The current substrate does not run Izhikevich cells on a continuous grid. Its Layer-1 unit is a **mass-spring cavity** — a damped oscillator coupled

through tunnels whose stiffness and damping are fixed by relation type. This is a different micro-model presenting the same macro-interface, which is precisely why the macro-layer transferred without modification.

Why the substitution happened. The spiking micro-layer was specified for biological fidelity, and biological fidelity turned out not to be load-bearing for the structural reasoning the engine is asked to do. The mass-spring cavity is simpler, has an analytic impedance, and admits a material table letting relation *type* set transmission properties. Nothing in §3 or §4 depended on the micro-model being spiking.

The general lesson is worth stating: **the layered specification survived a total replacement of its bottom layer.** That is evidence the interface between the two scales was drawn in the right place, and it is the most useful thing this paper produced.

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7. Limitations

Attractor convergence is assumed, not guaranteed. Section 4 asserts that the field settles. A driven nonlinear system need not converge; it may oscillate indefinitely, or lock into a ground state and refuse to leave. Ground-state locking was subsequently observed as a real failure mode and required a dedicated escape mechanism.

The decoder is unspecified. Stage 4 defers the conversion of a settled topology into text to "a downstream decoder", which is exactly where the irreducible translation cost of the architecture lives. Deferring it is not solving it, and naming it as deferred is the minimum honest treatment.

No account of acquisition. Constraints are stated. How a system acquires new constraints from experience lies outside this proposal and was addressed separately by local Hebbian consolidation.

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8. Conclusion

HNCA's contribution is the two-scale decomposition, and the specific claim that differential natural frequency suffices for domain separation without any routing logic. That claim survived into a substrate whose micro-layer bears no resemblance to the one proposed here — which is the strongest available evidence that the decomposition, rather than either individual layer, was the durable part.

References

- Izhikevich, E. M. *Simple Model of Spiking Neurons*, IEEE Transactions on Neural Networks, 2003.
- Kuramoto, Y. *Chemical Oscillations, Waves, and Turbulence*, 1984.
- Mahir, I. *From Artificial to Synthetic Intelligence.* §5.4c — the substrate in which the macro-layer runs.
- Mahir, I. *Voltage-Gated Harmonic Topology (VGHT).* — develops §3's frequency-separation argument into an explicit sub-band ladder.
- Implementation: `src/physics.hpp` (`struct MacroNode`, Kuramoto pass).